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Problem Chosen

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2004 Mathematical Contest in Modeling (MCM) Summary Sheet

(Attach a copy of this page to each copy of your solution paper.)

Type a summary of your results on this page. Do not include the name of your school, advisor, or team members on this page.

Summary

QuickPass is a clever way both to address the number one complaint that people have about Disney Parks—those long queues, and also to encourage visitors to spend more money on eating and shopping when they wander around, which those long queues had ever kept them from doing it.

After introducing Disney's scheme in use at present, we propose our scheme that is a mixture of reservation and flux regulation system, which could turn a torrential flux in peak hours to a smooth one.

The two schemes are tested in our stochastic simulation model in which we construct a psychological function about the probability of choosing QuickPass. We make comparison of regular and QuickPass queue according to two schemes based on wait time, queue length and the delay of the return time. It proves that in our scheme the average wait time of QuickPass is much shorter than that of Disney's when the arrival rate λ exceeds 4 people per minute.

Besides, we obtain a critical point of λ to decide when to open QuickPass system.

Then, further improvements are introduced, including an Automated Queue Regulation System and a multiple QuickPasses system in order to improve the utilization of the attractions in amusement park.

Based on the schemes above, we submit a non-technical summary and give our recommendation to park executives as a consultant.

Magic Regulation Scheme for QuickPass System

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1 Introduction

After exploring the concept of the QuickPass system in Disneyland Park, we introduce Disney's scheme in use at present. Considering the two possible anomalies, we analyze the causes of their occurrence and then propose our scheme to avoid them. We expatiate on the idea behind this scheme and analyze its difference with the former one.

Then, a stochastic simulation model is built to test the two schemes in which we construct a psychological function about the probability of choosing QuickPass. We make comparison of two schemes based on three criteria that the park executive cares about most, and give their respective strengths and weaknesses.

After that, we develop a second scheme referred to as an Automated Queue Regulation System (i.e.AQRS).

Finally, we bring forward the scheme of a multiple QuickPasses system in order to improve the utilization of the attractions in the amusement park.

2 A brief description of the QuickPass system

2.1 What is QuickPass?

"QuickPass" system is a free ride reservation system that gives you a ticket to ride a popular attraction during a specific time frame printed on your ticket. The system frees guests to do other things such as visiting other attractions, shopping or having a meal instead of waiting in queue while a ride is reserved for you. QuickPass is optional -- you can also wait in the regular queue if you want.

2.2 How the QuickPass system works?

There are two entrances for an attraction if a QuickPass system is offered---the regular one and the QuickPass one. Two posted wait time hangs on the respective entrance: one is for standby (regular queue), telling you that how long it will take if you go ahead to wait in regular queue, whereas the other is for QuickPass (virtual queue), telling you what time you will be able to ride the attraction through the QuickPass entrance if you get a QuickPass right now.

You can choose from two queues: regular or QuickPass. If the standby time exceeds to the amount of time you are willing to wait, you can go to a QuickPass machine nearby to get a QuickPass slip that indicates your return time. You are told to return at the designated return time and go for the attraction through the QuickPass entrance. Then you can go around for pleasures instead of waiting in queue before your turn to ride.

Besides, you had better be careful about your daily park entrance ticket, i.e. the admission ticket you used to get into the park. Because there is a bar code on the ticket that is used for the QuickPass slip to be issued to you. Each park entrance ticket has a unique bar code on it.

3 Problem Restatements

Consider an attraction with a QuickPass system.

Our jobs are:

- 1、 Propose schemes for the park executives to decide the return time of the customer if he gets a QuickPass slip.
- 2、 Determine the evaluating criteria of the schemes.

Our goals are:

- 1、 Increase the visitor's enjoyment of the amusement park.

What customers expect include that:

- The length of the QuickPass queue when they return is as short as possible.
- People's waiting time in QuickPass queue is as short as possible.
- They are eager to ride the popular attraction, i.e. the return time arranged by QuickPass system is as soon as possible.

2、 Help the executive to choose the QuickPass scheme.

What executives care about is:

- More visitors consume in the park
- Increase the utilization rate of facilities

3、 Avoid the anomalies that customers complained about.

Anomaly A: One comes first, but is not arranged first. For example, at a certain time QuickPasses were being offered for a return time as long as 4 hours later, whereas a short time later on the same ride, the QuickPasses were given for times only an hour or so later. It is unfair for the earlier customers.

Anomaly B: QuickPass is not faster than regular queue. The queue for people with QuickPasses is nearly as long and slow as the regular one.

4 Assumptions & Hypotheses

- 1、 **QuickPass system is available in peak hour.** That is to say, there is a high attendance for the attraction at that time. And the entertainment establishments in the attraction are busy.
- 2、 **Once the customers make their decision, it is unchangeable.**
- 3、 **The reserved time printed on the ticket lasts one-hour long.** This is for the convenience of people playing in the amusement park. They have more freedom to arrange their time for other entertainment, shopping or so, and be able to return in time.
- 4、 **The QuickPass slips and the posted wait time are updated every five minutes.** That is to say, the precision of the reserved time is 5 minutes. Consider the workload of changing the posted time signs on the entrances and the randomness of people's arrivals; at the same time it is impractical and unnecessary to change the time signs too frequently.
- 5、 **QuickPass holders are only allowed to ride the attraction during the designated time.**
- 6、 **The customers return to the QuickPass entrance during the first 45 minutes to insure a timely return.**
- 7、 **QuickPass queue and the regular queue do not interact.**
- 8、 **Each person can only have one active QuickPass reservation at a time.** The QuickPass system will not issue another ride reservation if you are already holding one. You must wait until after the start of your reserved time. From a perspective of the park executives, the idea behind this is to keep you in the park (and hopefully spending your money) as long as possible. And this policy may be changed in new QuickPass system.

5 Terms and Definitions

Table 1: terms and definitions

Symbol	Unit	Definition
i	—	The i st customer
j	—	The j st QuickPass holder
θ	min	The arrival time interval between two adjacent arrivals
	1/min	

λ		The arrival rate, showing λ people arrive per minute on average.
ν	1/min	The dequeue rate, showing ν people dequeue per minute on average
t	min	The arrival time
t_w	min	The posted wait time if you choose to wait in regular queue
T_w	min	The earliest designated time of reserved one-hour window for QuickPass
Δt	min	The delay of the play time
p	—	The probability of choosing QuickPass.
N	—	The number of QuickPasses issued for the same return time
M	—	The number of people simulated in our model

6 A queue system

A queue system is made up of three parts: enter a queue, queue and dequeue.

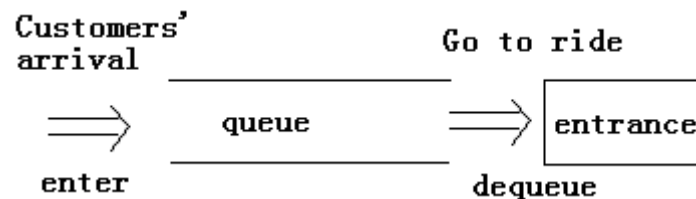


Figure 1: three parts of a queue

Input process (Poisson Process): The arrivals of the customers satisfy the three conditions of Poisson Process.

- The probability of arriving k customers during the time interval $[t, t + \Delta t]$ is related with Δt rather than t ;
- Non-aftereffect: The numbers of customers arriving at the time intervals that do not intersect are independent.
- During any finite time interval the number of customers arriving is finite.

According to these, it is rational to determine the customers' arrivals to be a Poisson Process. The time intervals between the adjacent arrivals follow the Exponential Distribution. And the distribution density function of θ is

$$f(\theta = x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0 \\ 0, & x \leq 0 \end{cases} \quad (1)$$

where,

t_i is the time the i st customer arrives.

$\theta_i = t_i - t_{i-1}$ stands for the time interval between the arrivals of the $i-1$ st and i st customers.

λ is the arrival rate. λ people arrive per minute on average.

$\frac{1}{\lambda}$ is mathematical expectation of the arrival interval θ .

Queue: People queue following the principle of “First come, first serve”.

Dequeue: Visitors leave the waiting queue to ride the attraction

ν is the rate of people who dequeue on average. The rate can be obtained by the former statistics of this attraction.

When $\frac{\lambda}{\nu} > 1$, i.e.